

AYE
TECH HUB

• REVIT BIM • LESSON 04

Levels and Grids in Practice

Build the Complete Framework for a 5-Storey Project

Lesson 04 of 30 | Revit BIM Program

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- Step-by-step: build 7 levels for a 5-storey commercial building
- Draw a 5×4 structural grid at 6m bay spacing in under 10 minutes
- Importing a DWG site plan and aligning Revit to true north
- Shared coordinates: linking the structural and architectural models
- Critical checks before you start modelling walls and columns

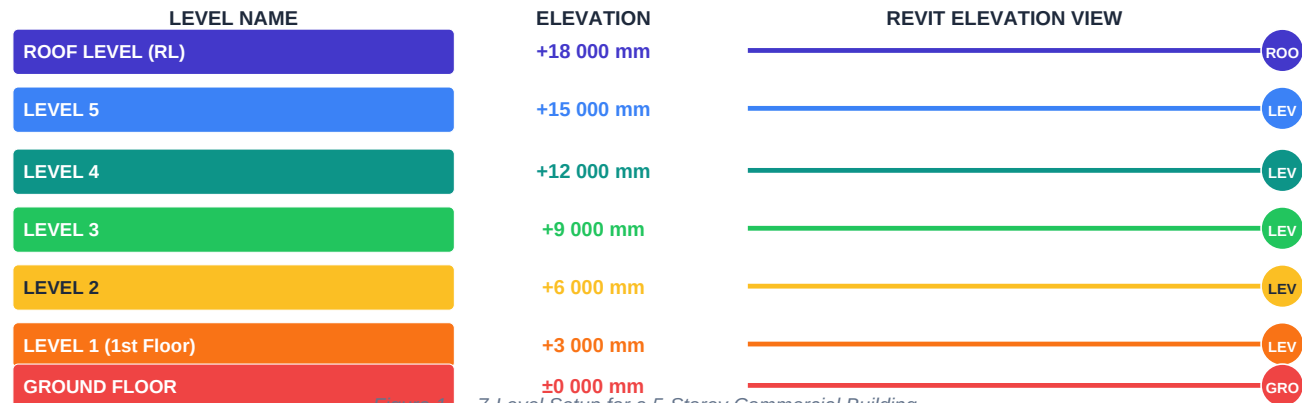
The Project: A 5-Storey Commercial Building

In this lesson we build the complete structural framework for a real project — a 5-storey commercial building. This is the work you do before placing a single wall or column. By the end of this lesson you will have: 7 levels (Ground Floor to Roof), a 5×4 structural grid at 6m bay spacing, the site DWG imported and aligned, and shared coordinates set for multi-discipline linking.

Project Brief

Parameter	Value	Notes
Building type	Commercial office	5 usable floors + roof plant
Ground floor use	Retail / reception	Higher ceiling: 4 500mm floor-to-floor
Typical office floors (1-4)	Open-plan office	3 600mm floor-to-floor
Roof level	Plant room + terrace	No habitable space
Structural grid	5 × 4 bays at 6 000mm	24 columns in the grid
Site orientation	True North = 15° from project north	From site survey DWG
Project units	Millimetres (mm)	Set before any modelling begins

Step 1 — Setting Up the 7 Levels



Detailed Level Setup Procedure

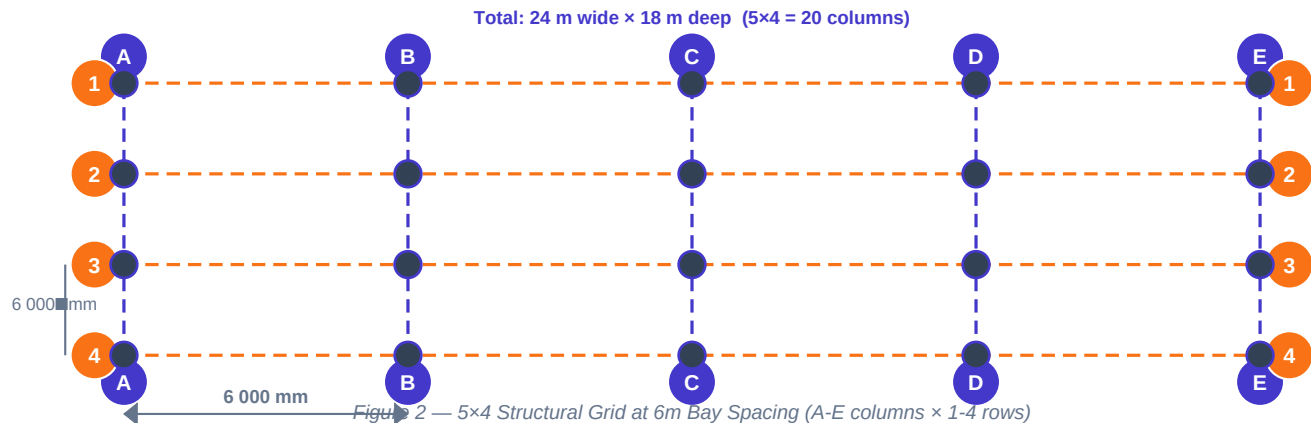
Action	Revit Steps	Key Check
1. Open an elevation view	Project Browser → Elevations → double-click "South Elevation"	Levels can ONLY be created and edited in Elevation or Section views
2. Create Ground Floor level	Architecture tab → Datum → Level → click at elevation 0 (already exists as "Level 1") → Rename to "Ground Floor" → set elevation = 0	This is your datum — everything is measured from here

3. Create Level 1 (1st Floor)	Select "Ground Floor" level → Copy → Constrain → type 4500 → Name: "Level 1" → Create associated plan view: YES	4500mm for retail ground floor; confirm with structural engineer
4. Create Levels 2-4 (office floors)	Select Level 1 → Copy → type 3600 for each (4 copies with offset) → Name: Level 2, 3, 4, 5 → Create plan views: YES	3600mm = comfortable office storey height with services void
5. Create Roof Level	Select Level 5 → Copy → type 3000 offset → Name: "Roof Level" → Create plan view: YES (for roof plan)	Plant room height — check equipment clearance requirements
6. Verify all elevations	Click each level head → confirm elevation shown is correct Total building height = 18 000mm from Ground Floor to Roof Level	Ground Floor: 0 L1: 4500 L2: 8100 L3: 11700 L4: 15300 L5: 18900 RL: 18000
7. Pin all levels	Select all levels → Pin (shortcut: PP)	Prevents accidental movement — critical once structural model is built on top

Why create associated plan views? When Revit asks "Do you want to create associated plan views?" — always say YES. This automatically creates a Floor Plan and Reflected Ceiling Plan for that level in your Project Browser. If you say No, you have to create them manually later.

Step 2 — Drawing the 5x4 Structural Grid

With levels complete, switch to the Ground Floor plan view and draw the structural grid. Our building has a regular 6m × 6m bay grid with 5 column lines across (A-E) and 4 column lines deep (1-4). This gives 20 column positions and 12 structural bays.



Step	Action	Revit Method
1	Open Ground Floor plan view	Project Browser → Floor Plans → double-click "Ground Floor"
2	Draw first vertical grid (A)	Architecture/Structure tab → Datum → Grid → click start point → click end point Line appears with bubble at top
3	Name grid "A"	Click the grid head bubble → type "A" → press Enter
4	Copy A to create B, C, D, E	Select grid A → Copy (shortcut: CO) → uncheck Constrain → Type 6000 → click to place B → type 6000 for C, D, E
5	Draw first horizontal grid (4)	Draw a horizontal grid line at a convenient Y position → name "4"
6	Copy grid 4 to create 3, 2, 1	Select grid 4 → Copy upward → 6000 offset each time → name 3, 2, 1
7	Make all grids 3D	Select all grids (Ctrl+A with grids visible) → Properties → check "3D" 3D grids appear in all views and linked models
8	Propagate extents to all views	Select all grids → right-click → Propagate Extents → select all floor plan views Grids appear at the same length in every plan view
9	Pin all grids	Select all grids → Pin (PP) → grids are locked from accidental move

Grid naming convention: In most countries, horizontal grid lines are numbered (1, 2, 3, 4...) and vertical grid lines are lettered (A, B, C, D...). Always match your grid names to the structural engineer's drawings. If the SE uses "1A" and "1B", use the same — grid coordination is critical for multi-discipline BIM.

Importing the Site DWG and Setting Coordinates

Most real projects start with a site survey drawing (DWG or DXF format) from the civil engineer or surveyor. Importing this into Revit and aligning the model to it ensures that the building sits correctly on the site — essential for coordination with civils, landscaping, and utilities.

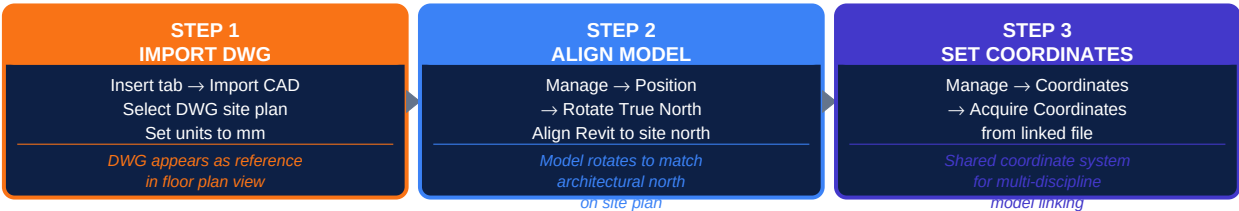


Figure 3 — DWG Import and Coordinate Alignment Workflow

Coordinate Concept	Definition	Why It Matters
Project Base Point (PBP)	The origin (0,0,0) of the Revit coordinate system. Fixed relative to the building.	All Revit measurements reference this point internally
Survey Point	The real-world reference point linked to the site survey coordinate system.	Allows the building to be positioned at its correct GPS / grid reference on Earth
Shared Coordinates	A common coordinate system shared between linked Revit files (Arch + Structural + MEP).	Ensures all three discipline models snap together correctly when linked
True North vs Project North	Project North = up on screen. True North = actual compass north. Often different.	Rotate True North to match site orientation for sun studies and reports
Acquire vs Publish Coordinates	Acquire: import coordinates from a linked file (survey DWG). Publish: share your coordinates to another linked Revit file.	The host model (usually Arch) acquires from survey, then publishes to Structural

Pre-Modelling Checklist — Before You Place the First Element

Check	How to Verify	If Wrong, Fix By
Units set to millimetres	Manage → Project Units → Length = mm	Change immediately — changing later causes dimension chaos
All 7 levels created and named correctly	Open South Elevation → verify all levels at correct elevations	Click level head → retype elevation; rename if needed
All levels have associated floor plan views	Project Browser → Floor Plans → confirm Ground Floor, Level 1-5, Roof Level	View tab → Create → Floor Plan → select missing level
Grid lines drawn, named A-E and 1-4	Open Ground Floor plan → verify 5 vertical + 4 horizontal grids	Draw missing grids; rename by clicking bubble
All grids pinned	Select a grid → check if pin icon appears (locked)	Select all grids → PP to pin
All levels pinned	Open elevation → select a level → check pin icon	Select all levels → PP to pin
Site DWG imported and aligned	Open Ground Floor → verify DWG visible and grid aligns to DWG column centres	Manage → Position → Move to Project to realign

Shared coordinates set	Manage → Coordinates → Project Base Point/Survey Point visible and not coincident	Acquire coordinates from survey DWG (Manage → Coordinates → Acquire)
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What comes next — REVIT-005: Wall Creation — drawing architectural walls in Revit: wall types (basic/stacked/curtain), wall base and top constraints, interior vs exterior walls, wall joins, and creating a custom wall type with multiple layers.